

Vibe-IC — ALL Phases / Stages / Steps (v2.2.0, exhaustive)

Plugin 0.2.2. Complete enumeration of every phase, stage, and step. There are **two coexisting views**: **(A)** the runner ground-truth markers (`[N/15]` / `def step_*`) that the code actually prints, and **(B)** the 33-step Stage model (sign-off view). Both are listed in full below; the companion `CANONICAL_FLOW_v2.2.0.md` is the narrative. Source of truth = the runners.

0. Entry + orchestrator + pre-flight

ID	Item	Where
P0	Pre-flight (env / PDK / tool availability)	<code>mcp_server_health_check</code> , <code>eda_doctor</code>
—	Path A: NL prompt / dialogue → Phase 1	<code>phase1_doc_one_shot_runner.py</code>
—	Path B: vendor docs → Phase 1 (docs mode)	same
—	Orchestrator: Phase 1 → Phase 2(=2a+2b) → Analog (after P2) → Phase 3	<code>vibe_ic_one_shot_runner.py</code>

A. RUNNER-MARKER VIEW (ground truth)

A.1 Phase 1 — 15 main steps + every sub-step (`phase1_doc_one_shot_runner.py`)

Marker	Step
<code>[1/15]</code>	Text extraction (input/docs → input_doc; 2 MB scan cap, v0.1.91)
<code>[2/15]</code>	L1_DATASHEET
<code>[3/15]</code>	L2_FRS
<code>[4/15]</code>	L3_CMD_PROTOCOL
<code>[5/15]</code>	L4_REGMAP
<code>[6/15]</code>	L5_ADI_SPEC
<code>[7/15]</code>	L6_CONTROL_LOGIC
<code>[8/15]</code>	L7_TEST_DEBUG
<code>[9/15]</code>	L8_RTL_CONSTANTS
<code>[10/15]</code>	L9_INTEGRATION_SPEC
<code>[11/15]</code>	L10_TEST_CASES
<code>[12/15]</code>	L11_OTP_CONTENT
<code>[13/15]</code>	L12_BEHAVIORAL_SEQUENCES
<code>[14/15]</code>	L13_LAB_CALIBRATION
<code>[14b/15]</code>	L8_TIMING_WAVEFORM
<code>[14b2/15]</code>	L8 protocol-width extract (R19)
<code>[14b3/15]</code>	L8 encoding-table overlay (R41)
<code>[14b4/15]</code>	L6 FSM / control-logic overlay (R42)
<code>[14b5/15]</code>	L12 behavioral-sequences overlay (R43)
<code>[14b6/15]</code>	L17/L18/L8_TIMING/L9 batch synth (R46) — DEPRECATED → 14c3
<code>[14b7/15]</code>	L8_RTL_CONSTANTS universal protocol constants (R48)
<code>[14c/15]</code>	L14-L18 protocol spec extract
<code>[14c0/15]</code>	L9 integration_spec overlay (R40)

Marker	Step
[14c1/15]	L1 protocol metadata overlay (R23)
[14c1b/15]	L17 handshake_pairs overlay (R27)
[14c2/15]	L3 protocol mirror from L14-L18 (R21)
[14c3/15]	L17/L18/L8_TIMING/L9 batch synth (R46 relocated)
[14c4/15]	L1/L2/L6/L7/L12 universal protocol doc facts (R50)
[14c5/15]	L4/L5/L10/L14/L15 residual cleanup (R52)
[14d/15]	L19-L23 skeleton emit
[14e/15]	serial_peripheral_protocol class synth (R53/R54/R55) — 81-protocol detector→synth dispatch
[14e2/15]	bus_interconnect_protocol Tier-2 synth (TileLink/Wishbone/Avalon/OCP/AXI-Stream)
[14e3/15]	Universal packet/PDU L10↔L3 opcode-consistency sweep
[15/15]	Coverage / parity report

→ 34 distinct Phase-1 markers (15 main + 19 sub/overlay/dispatch).

A.2 Phase 2 — step functions (phase2_one_shot_runner.py , logical order)

#	Step	Note
1	step_phase1	re-run/ingest Phase 1 if needed
2	step_rig_topology_skeleton	topology scaffold
3	step_rtl_gen	WAIVE for rtl_gen=null ic_class → spec-to-rtl role; phase2_scaffold_gen.py emits top/regs/fsm/tb/soc_wrap/cocotb
4	step_full_stack_tb_gen	self-checking TB
5	step_reference_tb	reference-TB conformance (eco_loop ≤ 3 retries)
6	step_yosys_synth	gate-level synth
7	step_qsf_gen	FPGA project
8	step_sdc_gen	constraints
9	step_otp_image_check	OTP image (if any)
10	step_fpga_compile	FPGA build → .sof
11	step_fpga_burn	program board
12	step_usb_hid_tester_verify	host protocol-tester acceptance
13	step_emit_phase2_manifests	manifests
14	step_final_audit	aggregate audit
(hook)	step_phase3	chains into Phase 3 when run end-to-end

Surrounding gates: rtl_hygiene_lint --fix, spec_conformance_check, chip_top_gate_wrapper_gen, MCP_eda_lint / eda_synth / eda_cocotb.

A.3 Phase 3 — step functions (phase3_one_shot_runner.py)

#	Step	Tool (open-source)
1	step_synth	yosys (+ tie-cell pass: setundef -zero; hilomap; splitnets; clean)
2	step_pnr	

#	Step	Tool (open-source)
		OpenROAD (floorplan→PDN→place→CTS→route)
3	step_gds	KLayout (def2gds)
4	step_drc	KLayout sky130 deck
5	step_lvs	netgen / yosys_equiv — the LVS sign-off chain (S D)
6	step_canonicalize_artefacts	normalize outputs

A.4 Analog A1–A8 (analog_one_shot_runner.py, parallel to Phase 2)

Step	Name	Output
A1	spec_extract	analog/<block>/A1_spec.json
A2	topology_select	A2_topology.json
A3	netlist_gen	<block>.sp
A4	corner_sweep	A4_corners.json
A5	layout (Magic)	A5_layout.json (needs DRC-clean + LVS-match flags)
A6	post_layout_resim	A6_postsim.json
A7	hardmacro_gen	{.lef, .lib, .gds, .v} → feeds Phase 3
A8	hw_verify (HIL)	A8_hw_verify.json

A.5 Mixed-signal M1–M4 (skill-level, no dedicated runner)

Step	Name	Where
M1	top merge	mixed_signal_m1_top_merge_check.py + skill mixed-signal-cosim
M2	co-sim setup	skill mixed-signal-cosim
M3	co-sim run	skill mixed-signal-cosim
M4	integration verify	skill mixed-signal-cosim

B. 33-STEP STAGE MODEL (sign-off view, 33_step_flow_overview.md)

Stage 1 — RTL Generation + Verification (steps 1-6)

#	Step	Tool/Skill	Gate
1	Spec-to-RTL	spec-to-rtl	L1-L9 present + RTL emitted
2	Lint	eda_lint + Phase-2a gates + polluter check	Verilator 0 errors
3	CDC / RDC check	cdc-check	CDC paths synchronised
4	Simulation	testbench-gen + eda_simulate + coverage	all tb PASS + coverage
5	Formal verification	formal-verify + assertion- gen	k-induction proved
6	FPGA early prototype	fpga-test-harness + eda_fpga_compile/program + on-board BIST	.sof + BIST PASS

Stage 2 — Synthesis + DFT (steps 7-13)

#	Step	Tool/Skill	Gate
7	Constraint setup	constraint-gen	*.sdc + 3-corner pvt_matrix.json
8	SDC validation	SDC lint	all clock/IO constraints present
9	Synthesis (Yosys)	eda_synth + synth-doctor	mapped netlist + cell count
10	Pre-layout STA	eda_sta_mcorner (SS/TT/FF)	WNS/WHS all corners
11	DFT insertion (scan + ATPG)	dft-insert + atpg + eda_dft	scan chain + stuck-at ≥ 85%
12	Post-DFT optimization	resynth / buffering	timing held
13	Equivalence check	equivalence-check + Yosys equiv	RTL ≡ post-DFT netlist

Stage 3 — Physical Design + Sign-off (steps 14-28)

#	Step	Tool/Skill	Gate
14	Floorplan + PDN	eda_pnr (init)	area / utilization
15	Clock planning	clock-planning skill	clock-buffer list
16	Placement (global + detailed)	eda_pnr	placement legal
17	CTS	eda_pnr enable_cts=true	clock skew
18	Post-CTS hold fixing	repair_timing -hold	WHS > 0
19	Routing (global + detailed)	eda_pnr enable_detailed_route=true + def_stage_progression_check	0 overflow + DEF SHA differs
20	Parasitic Extraction (RC→SPEF)	eda_extraction	spef_extraction_check PASS
21	Post-route STA (MMMC)	eda_sta_mcorner	3-corner pass
22	IR Drop	eda_ir_drop	static + dynamic
23	EM check	electromigration	lifetime ≥ 10 yr
24	Antenna check	OpenROAD antenna	0 violation
25	Signal Integrity (crosstalk/noise)	SI analysis	si_crosstalk_check PASS
26	Post-Layout Gate-Level Sim (+SDF)	eda_simulate	post_layout_sim_check PASS
27	Physical Verification	eda_drc_klayout + eda_lvs + ERC + Density	DRC=0 / LVS match / ERC=0
28	ECO repair loop	eco-plan	eco_loop_audit PASS

Stage 4 — Output + Validation (steps 29-33)

#	Step	Tool/Skill	Gate
29	Power analysis	pre + post layout	meets spec
30	Metal Fill (density fill)	eda_pnr	metal_fill_density_check PASS
31	Tapeout checklist	tapeout-checklist + signoff_audit	4/4 strict
32	GDSII output	eda_gds + def2gds	only if step 27 clean
33	FPGA final sign-off	recompile + on-board test + fpga_on_board_attestation_ch eck	bitstream hash + hw evidence

Numbering caveat: the Stage-3 sign-off checks have a SECOND numbering in the sign-off-audit scheme used in § D (SPEF 22 / STA 23 / IR 24 / EM 25 / Antenna 26 / SI 27 / DRC-LVS-ERC 30 / fill 33), which differs from this 33-step doc (SPEF 20 / STA 21 / IR 22 / EM 23 / Antenna 24 / SI 25 / PV 27 / fill 30). To be unified by `flow_doc_emit.py` (below).

C. The LVS sign-off chain (under Phase-3 `step_lvs`, NEW v0.1.96→v0.2.2)

Layer	What	Tool
1	Structural LEC (default)	<code>eda_lvs mode=yosys_equiv</code> (equiv_simple + equiv_induct) — SAT-model unproven = Category-D gap
2	Device-level coverage	<code>eda_extraction</code> (magic ext2spice) + <code>eda_lvs mode=netgen</code> + <code>lvs_netgen_setup_emit.py</code>
3	Powered-netlist closure	OpenROAD <code>write_verilog -include_pwr_gnd</code> (after <code>global_connect</code>)
4	Top-level port labels	Route A <code>magic_port_extract_emit.py</code> (port makeall) / Route B <code>lvs_def_port_seed.py</code> (DEF seed)
5	Sign-off guard (MANDATORY)	<code>lvs_signoff_guard.py</code> — RAISES on a portless / vacuumous match

D. Phase-3 sign-off checks — gap status (sign-off-audit numbering)

Step	Check	Open-source status	Severity
22	SPEF (OpenRCX)	<code>extract.tcl</code> must <code>global_route + set_wire_rc</code> first	◆ medium
23	Post-route STA (MMMC)	runs once SPEF exists; pilots report slack +X ns MET	● none (passes)
24	IR drop (PSM)	cascading-missing on SPEF	◆ medium
25	EM	cascading-missing on SPEF	◆ medium
26	Antenna	router runs it; report not on audit path	● low
27	SI (crosstalk)	cascading-missing on SPEF	◆ medium
30	DRC / LVS / ERC	sky130 ships only Calibre decks; open decks need wiring (LVS chain = § C)	● high (env)
33	Metal fill	runner lacks the fill stage → no <code>filled.def</code>	◆ medium
18	Spare cells	30 placed; <code>spare_cells.json</code> missing <code>rows[]</code>	● low (schema)
5	Formal	<code>altsyncram</code> no formal model → INFORMATIONAL waiver	● none

None is a circuit-design error — all script-ordering / cascading / environment / report-schema. Actionable fixes tracked in `ORGANIC-20260531-phase3-signoff-chain-open-source-gaps`.

E. Totals

View	Count
Phase 1 runner markers	34 (15 main + 19 sub)
Phase 2 step functions	14 (+1 phase3 hook)
Phase 3 step functions	6
Analog	A1-A8 (8)
Mixed-signal	M1-M4 (4)
33-step Stage model	33 (Stage 1:6 / 2:7 / 3:15 / 4:5)
LVS sign-off chain layers	5
Phase-3 sign-off checks	10

F. Keep current

This is a **derived** enumeration. Regenerate after a runner change: `grep -noE '\[[0-9]+\[a-z0-9_\]*/[0-9]+\][^"]*'` `phase1_doc_one_shot_runner.py`, `grep -noE 'def step_\[a-z0-9_\]+' phase{2,3}_one_shot_runner.py`, and `analog_one_shot_runner.py`'s A1-A8 header. Backlog candidate: `programs/flow_doc_emit.py` to emit this deterministically (and unify the two Stage-3 numberings) the way `INDEX.md` is generated.